

DIRECTED ENERGY WEAPONS:

Particle Beam Weapons Technology



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Part 1 of this article last month covered directed energy weapons, kinetic energy weapons and particle beam weapons. This part covers the involved technology areas and advantages as well as limitations of the particle beam weapons.

Different technology disciplines related to development of particle beam weapons include particle accelerator, power source driving the accelerator, target-tracking and beam-pointing system, beam propagation through medium separating the source and the target and lethality (Fig. 6).

Particle accelerator technology

The particle accelerator is the source of high-energy particle beam and is therefore the heart of the particle beam weapon system. Three main constituents of the particle accelerator are: the source of particles such as electrons, protons or charged atoms, the device that injects these particles into the accelerating section and finally the accelerating section itself.

The particle accelerator accelerates particles to extremely high energies. These particles, as outlined earlier, are elementary particles like electrons or protons, or even whole atoms. In modern accelerators the particles very quickly attain nearly the speed of light, which is about 3×10^8 km/s. According to Einstein's theory of relativity, this speed can only be reached by massless particles and can never be exceeded by any particle. Though the speed of a

particle cannot be increased any further after it has attained near speed of light, its kinetic energy can still be increased. All particle accelerators accelerate the particles roughly to the speed of light; difference between more or less powerful accelerators is the kinetic energy of the accelerated particles.

There are two types of accelerators, namely, the linear accelerators and the circular accelerators (also called ring accelerators). Linear accelerators accelerate particles over a long, straight line, where the particle beam travels from one end to the other. In circular accelerators, powerful magnets are used to bend the particle's path into a circle and the beam of particles travels repeatedly in a loop. Technologically, both linear and circular accelerators are in a highly advanced state of development. The preferred accelerator technology depends upon the application.

In earlier accelerators, strong electric fields were used to accelerate the particles, but the maximum usable electric field strength had an upper limit due to spontaneous discharge. The accelerating section of all conventional linear accelerators therefore is made up of a cascade arrangement of a number of accelerating segments that sequentially apply an accelerating electric field to the charged particles. While the voltage in each segment may be relatively low, the repeated application of an accelerating voltage by the large number of modules ultimately produces substantially high particle energies.

Modern accelerators use focusing magnets to ensure that particles traveling at the speed of light are not lost and always remain on their reference trajectory. SLAC's 3.22km

Fig. 6: Particle beam weapon technology areas

